

DAVID G. FOBAR

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FORMAL EDUCATION

University of Michigan <i>M.S. Nuclear Engineering</i>	Ann Arbor, MI 2018
Missouri University of Science and Technology <i>M.S. Engineering Management</i>	Rolla, MO 2014
United States Military Academy <i>B.S. Mechanical Engineering</i>	West Point, NY 2010

LEADERSHIP EXPERIENCE - US ARMY

Deputy Director <i>DTRA Nuclear Science and Engineering Research Center (NSERC)</i>	September 2022 – Present West Point, NY
Instructor/Assistant Professor <i>Department of Physics and Nuclear Engineering</i>	July 2018 – Present West Point, NY
Commander <i>HHC/85th Civil Affairs Brigade</i>	March 2015 – April 2016 Fort Hood, TX
BDE Assistant Plans Officer <i>36th Engineer Brigade</i>	Oct. 2014 – Feb. 2015 Fort Hood, TX/Liberia
S1/S3/XO SFAAT Advisor <i>1-101 Air Assault Division</i>	Nov. 2012 – Aug. 2013 RC-E, Afghanistan
Infantry Platoon Leader <i>C/1-32 Cavalry Regiment, 1-101 Air Assault Division,</i>	Jan. 2012 – Nov. 2012 Fort Campbell, KY

EDUCATOR EXPERIENCE - UNITED STATES MILITARY ACADEMY

Guest Lecturer: Strategic Nuclear Deterrence and Weapons Effects	AY24, AY25
• Introduction to Transportation Engineering - CE495 • Science and Technology in Policy - PH456 • Various ROTC detachments.	
Course Director/Instructor: Radiation Instrumentation and Shielding - NE452	AY22
Course Director/Instructor: Introduction to Nuclear Reactor Engineering - NE355	AY22
Laboratory Program Director: Core Physics (PH2x5, PH2x6)	AY20-21
Instructor: Physics I - PH205, PH255	AY19, AY20-2
Instructor: Physics II - PH206, PH256	AY20-1, AY21
Advisor: Capstone and Research Advisor (multiple levels within PANE, EECS, CME, XE)	AY19 - AY25
• Probabilistic Volumetric Point Source Localization Using Recursive OctoTree Sub-Voxelization and Kalman Filtering • SCRAM - Scatterable Radiation Monitoring • RADBOT/E2MAPs - Semi-autonomous radiation detection robot	

- Machine learning enabled gamma-ray and neutron particle discrimination
- **SP**ace **E**ngineering and **A**ppplied **R**esearch (SPEAR)
SPEAR initially aimed to reconstitute lost space-based assets.
- **O**utdoor **H**igh **E**nergy **L**aser **L**aboratory (O-HELL)
O-HELL allows for the safe conduct of high energy laser range operations at West Point, NY.

PEER REVIEWED PUBLICATIONS

Fobar D, Filip K, Vitale S, Koch W. "Silicon-on-Uranium-Oxide Substrate for Single Event Upset Evaluation", in Journal of Radiation Effects Research and Engineering, Volume 44.

Discusses a novel method of preparing microelectronics for radiation harness testing. The work is purely computational at this time and pending submission for patent review.

Halbert T, **Fobar D**. "Development of a Pulsed UV-LED Calibration Technique for Scintillating Detectors", in Journal of Radiation Effects Research and Engineering, Volume 44.

This work introduces a method of conducting source-less energy calibration of radiation detectors using a small pulsed LED.

Fobar D, Vanderlip W, Koch W, Chapman P. "A Machine Learning Approach to a Multidetector Array Response Function for Nuclear Search", in IEEE Transactions on Nuclear Science, Aug 2022.

This work investigated the novel use of machine learning to project a unit vector from an array of radiation detectors back to a point source.

Fobar D, Wilhelm AS, Phillips L, Chapman P. "Considerations for Training an Artificial Neural Network for Particle Type Identification", in IEEE Transactions on Nuclear Science, Sep 2021.

Machine learning was used for conducting pulse shape discrimination between neutron and gamma-ray radiation particles interacting in an organic scintillating radiation detector.

Fobar D, Xiao X, Burger M, Le Berre S, Motta, A, Jovanovic I. "Robotic delivery of laser-induced breakdown spectroscopy for sensitive chlorine measurement in dry cask storage systems", in Spectrochimica Acta Part B: Atomic Spectroscopy, November 2018

A method of remotely characterizing chlorine deposition on the surface of stainless steel using dual-pulse LIBS was developed for the in-situ inspection of dry-cask storage canisters of spent nuclear fuel.

Xiao X, Le Berre S, **Fobar D**, Burger M, Skrodzki P, Hartig K, Motta A, Jovanovic I. "Measurement of chlorine concentration on steel surfaces via fiber-optic laser-induced breakdown spectroscopy in double-pulse configuration", in Spectrochimica Acta Part B: Atomic Spectroscopy, March 2018

A method of characterizing chlorine deposition on the surface of stainless steel using dual-pulse laser induced breakdown spectroscopy was developed.

CONFERENCE PRESENTATIONS AND PUBLICATIONS

Filip K, **Fobar D**, Ague S. "Reconstructing Nuclear Fallout Fields with Scatterable Dosimeters", 2024 IEEE Long Island Systems, Applications and Technology Conference.

We documented the wireless infrastructure and data processing necessary to localize the sensors using differential time-of-arrive and filter their signal using a Kalman filter.

Fobar D, Koch W, Bruff T, Chapman P. "Gamma ray directionality from a tightly packed scintillator array", presentation at SORMA XIX, MAY 2023.

This research supplements the RADBOT capstone and is an extension of the work published in Aug 2022. The ANN is trained using a physics informed loss function and includes probabilistic weighting of source location.

Koch W, **Fobar D**, Bruff T, Chapman P. "Demonstration of a Low Power Imaging System for a Portable Time Projection Chamber", presentation at HEART, APRIL 2023 by Koch W., pending publication in JRERE following peer-review.

This engineering project aims to provide its operator real-time localization of fast neutron sources - a significant indicator for special nuclear material.

Liebers N, Huckleberry J, Ruiz D, **Fobar D**, Chapman P. "Unsupervised Machine Learning Approaches to Nuclear Particle Type Classification", published in IEEE LISAT Conference Proceedings, May 2022.

As an extension to previous research, an algorithm using unsupervised machine learning was developed by cadets during a joint independent research project between EECS and PANE.

OTHER PUBLICATIONS

Fobar D, Halbert T. "The Imperative of Nuclear Literacy in Military Education Insights from the NSERC Nuclear Deterrence Staff Ride", in Countering-WMD, Fall/Winter 2025.

This article communicates the need and usefulness of educating the next generation of military leaders on the nuclear weapons enterprise. The article walks through the various sites visited and what they contribute to the cadets and midshipmens understanding. This event has demonstrated to be a life changing experience for the young leaders.

Filip K, **Fobar D**, Halbert T, Ague S. "Reconstructing Fallout Distributions: Using convolution Neural Networks to Reconstruct Nuclear Fallout Distributions from SCatterable RAdiation Monitors (SCRAM)", in Countering-WMD, Summer/Fall 2024.

This article discusses various data sources available for generating artificial radiation fields in support of training a neural network that is able to reconstruct a continuous radiation intensity map from randomly scattered point detectors.

Wilhelm AS, Phillips L, **Fobar D**. "Optimizing the Teaching of Technical Computing for C-WMD at West Point", in Countering-WMD, Summer/Fall 2020.

This article showcased the facilities available for computation instruction within the D/PANE. While the journal article is not considered peer-reviewed, it is shared across the Counter Weapons of Mass Destruction community and raised awareness of the facilities available. This resulted in other academic department requesting the facilities use.